

RYAN SOLOMON

SOFTWARE ENGINEER | FULL-STACK & FRONTEND ARCHITECTURE | AI APPLICATIONS

Knoxville, TN • Open to Remote • ryan@ryansolomon.io • (563) 608-3464 • linkedin.com/in/ryan-solomon-32b4121b3

SUMMARY

Software engineer with 5.5 years building and modernizing a production healthcare platform serving 1,000+ users. Primary contributor to the product frontend and shared component library, with progressively broader ownership across full-stack development, architecture, production AI, database performance, testing, security, and EHR integrations. Experienced leading large migrations on a live product, designing reusable platform capabilities, and diagnosing complex production failures.

EXPERIENCE

Unite Genomics | Software Engineer | Feb 2021 - Present

- Primary contributor to the product frontend, authoring 5,616 of 10,510 commits (53%) across a ~12-engineer team and 54% of a shared TypeScript component library; delivered 1,700+ tracked issues and 18 epics.
- Progressively expanded from frontend development into full-stack and architectural ownership across React, Django/DRF, PostgreSQL, Redis, and production application services.
- Led 12 major framework migrations on a live product without a feature freeze, including React 16→19, MUI v5→v9, Kendo React 9→14, Recoil→Jotai, SWR v1→v2, SurveyJS v1→v2, and Create React App/webpack→Vite; sequenced the state migration into seven independently shippable, revertible steps.
- Architected a database-served white-label theming platform that converted a single-brand product into a five-tenant platform from one codebase, with a theme-generation CLI and operations workflow that moved partner onboarding off the engineering backlog.
- Designed and published a reusable TypeScript/Kendo data-grid library used across 20+ tables, centralizing column behavior while moving pagination, sorting, and filtering server-side for large patient datasets.
- Shipped a production AI provider-search assistant over a 13,000-organization directory using the OpenAI Agents SDK, with SSE streaming, persisted conversation history, guardrails, per-user rate limits, and a circuit breaker; root-caused SQL ranking defects that had been attributed to model quality.
- Optimized a degrading admin-dashboard query from 185ms to 2.7ms in a 2,000-patient benchmark by converting joined aggregates to correlated subquery annotations; added regression coverage for query shape and validated a failed partial optimization that was ~10x slower than the original.
- Raised frontend test coverage from under 50% toward 75%+ overall and 90%+ on Account Settings across 481 specs; converted fixture-mocked Cypress tests to true end-to-end coverage and added flake detection, CI parallelization, and zero-tolerance console-error gating.
- Built an internal admin portal from scratch with 12 operational screens for provider curation, conflict review, audit history, patient notifications, and background-import monitoring, eliminating direct production database edits as the standard workflow.
- Delivered HIPAA Release of Records end to end and integrated FHIR R4/DSTU2 imports across Epic, Cerner, Athena Health, and HAPI, including vendor capability checks and structured upstream error classification.
- Introduced production observability with Sentry across React and Django, including source maps, release tagging, and PHI scrubbing; addressed token/PII logging, client-side eval, email amplification, and other security findings.
- Authored 11 technical design specifications in the final 18 months covering schema design, API contracts, migration sequencing, and end-to-end test strategy, documenting alternatives and tradeoffs.

SELECTED TECHNICAL STRENGTHS

Frontend: React 16-19, TypeScript, JavaScript/ES2022+, MUI 5-9, Kendo React 9-14, Jotai, Recoil, SWR, SurveyJS, Vite, responsive design, WCAG 2.1 AA

Backend & Data: Django, Django REST Framework, Python, PostgreSQL, Redis, REST APIs, SQL/query optimization, derived data models

AI & Integrations: OpenAI Agents SDK / ChatKit, SSE streaming, tool calling, prompt guardrails, rate limiting, circuit breakers, FHIR R4/DSTU2, EHR integrations

Testing & Delivery: Cypress, Jest, React Testing Library, Vitest, pytest, E2E testing, GitHub Actions, Git, Docker/docker-compose, Sentry

Cloud & Services: AWS Cognito, S3, SES, SQS, EKS; Postmark; Zapier

EDUCATION

M.A. in Kinesiology — University of Northern Iowa | Graduate GPA: 4.0